The second–derivative lower–bound function (SeLF) algorithm and three acceleration techniques for optimization with strongly stable convergence

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Abstract. In this paper, we propose a new maximization method, called as the *second–derivative lower–bound function* (SeLF) algorithm. The SeLF algorithm is a general principle for iteratively calculating the MLE $\hat{\theta}$ of the parameter θ in a one-dimensional target function (usually, the marginal log-likelihood function) $\ell(\theta)$ and its each iteration consists of two steps: A second–derivative lower–bound function step (SeLF-step) and a maximization step (M-step), where the SeLF-step firstly finds a function $b(\theta)$ satisfying $\ell''(\theta) \geqslant b(\theta)$ for all $\theta \in \Theta$ and secondly constructs a surrogate function $Q(\theta|\theta^{(t)})$ [whose form depends on $\theta^{(t)}$ being the t-th iteration of $\hat{\theta}$] minorizing $\ell(\theta)$ at $\theta = \theta^{(t)}$, and the M-step calculates the maximizer $\theta^{(t+1)}$ of the $Q(\theta|\theta^{(t)})$ function, which is equivalent to solving the equation $\ell'(\theta) + \int_{\theta^{(t)}}^{\theta} b(z)\,\mathrm{d}z = 0$ to obtain its explicit solution $\theta^{(t+1)}$. The SelF algorithm holds two major advantages: (i) It strongly stably converges to the MLE $\hat{\theta}$, in contrast to general *minorization–maximization* (MM) algorithms only possessing weakly stable convergence; and (ii) it does not depend on any initial values, in contrast to Newton's method. The key for applying the SeLF algorithm is to find a function $b(\theta)$ satisfying $\ell''(\theta) \geqslant b(\theta)$ for all $\theta \in \Theta$ such that an explicit solution to the equation $\ell'(\theta) + \int_{\theta^{(t)}}^{\theta} b(z)\,\mathrm{d}z = 0$ is available. Furthermore, we develop three acceleration techniques (i.e., optimal SeLF, sub-optimal SeLF, and fast-SeLF algorithms) for the SeLF algorithm, resulting in a weakly stable convergence. Various applications in statistics of the proposed SeLF algorithm and three acceleration versions are introduced. The analysis of the convergence rates of these algorithms are provided. Some numerical experiments and comparisons are also presented.

Keywords: Acceleration technique; Convergence analysis; MM algorithms; SeLF algorithm; Strongly/weakly stable convergence.

1. Introduction

Suppose that $f(x|\theta)$ is a *probability density function* (pdf) with parameter $\theta \in \Theta$ and let $Y_{\text{obs}} \triangleq \{x_i\}_{i=1}^n$ be a realization of $\{X_i\}_{i=1}^n \overset{\text{iid}}{\sim} f(x|\theta)$. The log-likelihood function of θ is

$$\ell(\theta) = \sum_{i=1}^{n} \log[f(x_i|\theta)]. \tag{1.1}$$

For many statistical models, it is difficult to calculate the MLE of θ from the log-likelihood function because there exists no explicit expression for the $\hat{\theta}$. To handle this problem, many numerical methods have been put forward, like, *Newton–Raphson* (NR) algorithm, gradient ascent algorithm, MM algorithm and so on forth.

1.1 Drawbacks of traditional numerical algorithm

1) Although the NR algorithm enjoys quadratic convergence rate, the NR iteration may tend to different points if we choose the wrong initial values;

2) It is difficult to determine the step size for the gradient ascent algorithm because it may be time-consuming for a small step size and it may diverge for a large step size;

3) The MM iteration enjoys uphill property so that it is able to achieve the maximizer. The difficulty of the MM algorithm is that we can not always find a proper Q function minorizing the log-likelihood function.

1.2 The MM Algorithms

Assume that $Q(\boldsymbol{\theta}|\boldsymbol{\theta}^{(t)})$ minorizes $\ell(\boldsymbol{\theta}) \triangleq \ell(\boldsymbol{\theta}|Y_{\text{obs}})$ at $\boldsymbol{\theta} = \boldsymbol{\theta}^{(t)}$, i.e.,

$$\begin{aligned} Q(\boldsymbol{\theta}|\boldsymbol{\theta}^{(t)}) &\leqslant \ell(\boldsymbol{\theta}), \quad \forall \boldsymbol{\theta},\ \boldsymbol{\theta}^{(t)} \in \boldsymbol{\Theta} \quad \text{and} \\ Q(\boldsymbol{\theta}^{(t)}|\boldsymbol{\theta}^{(t)}) &= \ell(\boldsymbol{\theta}^{(t)}). \end{aligned}$$

If we could find such a real value function $Q(\boldsymbol{\theta}|\boldsymbol{\theta}^{(t)})$ depending on $\boldsymbol{\theta}, \boldsymbol{\theta}^{(t)} \in \boldsymbol{\Theta}$, where $\boldsymbol{\theta}^{(t)}$ denotes the t-th approximation of the MLE $\hat{\boldsymbol{\theta}}$, then by maximizing $Q(\boldsymbol{\theta}|\boldsymbol{\theta}^{(t)})$ instead of the

target log-likelihood function $\ell(\boldsymbol{\theta}|Y_{\text{obs}})$, we obtain the maximizer of $Q(\boldsymbol{\theta}|\boldsymbol{\theta}^{(t)})$ as

$$\boldsymbol{\theta}^{(t+1)} = \arg\max_{\boldsymbol{\theta}\in\boldsymbol{\Theta}} Q(\boldsymbol{\theta}|\boldsymbol{\theta}^{(t)}). \tag{1.2}$$

The ascent property of the MM algorithm. Let $\boldsymbol{\theta}^{(t+1)}$ be defined in (1.2), then we have

$$\ell(\boldsymbol{\theta}^{(t+1)}) \geqslant Q(\boldsymbol{\theta}^{(t+1)}|\boldsymbol{\theta}^{(t)}) \geqslant Q(\boldsymbol{\theta}^{(t)}|\boldsymbol{\theta}^{(t)}) = \ell(\boldsymbol{\theta}^{(t)}).$$

An increase in $Q(\boldsymbol{\theta}|\boldsymbol{\theta}^{(t)})$ forces an increase in $\ell(\boldsymbol{\theta})$. This ascent property guarantees a monotone convergence of an MM algorithm.

2. The SeLF algorithm with strongly stable convergence

A sequence $\{\theta^{(t)}\}_{t=0}^{\infty}$ is called to weakly stably converge to a fixed point θ^*, if

$$|\theta^{(t+1)} - \theta^*| < |\theta^{(t)} - \theta^*|, \qquad \forall\, t = 0, 1, \ldots, \infty.$$

A sequence $\{\theta^{(t)}\}_{t=0}^{\infty}$ is called to strongly stably converge to a fixed point θ^*, if

$$\theta^{(0)} < \theta^{(1)} < \cdots < \theta^{(t)} < \cdots \leqslant \theta^* \qquad \text{or} \qquad \theta^* \leqslant \cdots < \theta^{(t)} < \cdots < \theta^{(1)} < \theta^{(0)}.$$

An algorithm is said to possess *weakly (or strongly) stable convergence* if the sequence $\{\theta^{(t)}\}_{t=0}^{\infty}$ generated by this algorithm weakly (or strongly) stably converges to a fixed point θ^*. It is clear that a strongly stable convergence implies a weakly stable convergence.

2.1 Definition of a SeLF

Let $\ell(\theta)$ denote a one–dimensional target function (usually, the log-likelihood function of the parameter θ) of interest, and its first– and second–order derivatives exist, denoted by $\ell'(\cdot)$ and $\ell''(\cdot)$, respectively. Assume that we are interested in the maximization of $\ell(\theta)$ to obtain the MLE of θ:

$$\hat{\theta} = \arg\max_{\theta\in\Theta} \ell(\theta), \tag{2.1}$$

where $\Theta \subseteq \mathbb{R} \triangleq (-\infty, \infty)$ is the parameter space. Suppose that there exists a function $b(\theta)$ also defining on Θ such that

$$\ell''(\theta) \geqslant b(\theta), \quad \forall\, \theta \in \Theta, \tag{2.2}$$

where $b(\theta)$ is called the *second–derivative lower–bound function* (SeLF) for $\ell(\theta)$.

The basic idea behind the SeLF algorithm is transferring the difficult maximization in (2.1) to a tractable maximization problem by constructing a minorizing function (denoted by $Q(\theta|\theta^{(t)})$ with $\theta^{(t)}$ being the t-th approximation of $\hat{\theta}$) based on the SeLF of $\ell(\theta)$. Thus, the key to the SeLF algorithm is to find a proper SeLF satisfying the following two conditions:

(i) The maximizer $\theta^{(t+1)}$ of the minorizing function $Q(\theta|\theta^{(t)})$ can be calculated analytically.

(ii) The SeLF $b(\cdot)$ should be tight to the $\ell''(\cdot)$.

The condition (i) aims to obtain a closed–form expression for $\theta^{(t+1)}$, while the condition (ii) plays a key role in accelerating the SeLF algorithm, which will be discussed in Section 4.

2.2 Construction of a minorizing function

To construct a minorizing function, based on the SeLF $b(\theta)$ satisfying (2.2), we first define

$$B(\theta|\theta^{(t)}) \triangleq \int_{\theta^{(t)}}^{\theta} b(z)\,\mathrm{d}z, \tag{2.3}$$

and then we introduce the following auxiliary function

$$h(\theta|\theta^{(t)}) \triangleq \ell(\theta) - \int_{\theta^{(t)}}^{\theta} B(z|\theta^{(t)})\,\mathrm{d}z. \tag{2.4}$$

The first– and second–order derivatives of $h(\theta|\theta^{(t)})$ with respect to θ are given by

$$h'(\theta|\theta^{(t)}) = \ell'(\theta) - B(\theta|\theta^{(t)}) \quad \text{and} \quad h''(\theta|\theta^{(t)}) = \ell''(\theta) - b(\theta) \overset{(2.2)}{\geqslant} 0, \tag{2.5}$$

respectively, so that

$$h(\theta^{(t)}|\theta^{(t)}) \overset{(2.4)}{=} \ell(\theta^{(t)}) \quad \text{and} \quad h'(\theta^{(t)}|\theta^{(t)}) \overset{(2.5)}{=} \ell'(\theta^{(t)}) - B(\theta^{(t)}|\theta^{(t)}) \overset{(2.3)}{=} \ell'(\theta^{(t)}). \tag{2.6}$$

By using the supporting hyperplane inequality, we have

$$h(\theta|\theta^{(t)}) \overset{(2.5)}{\geqslant} h(\theta^{(t)}|\theta^{(t)}) + h'(\theta^{(t)}|\theta^{(t)})(\theta - \theta^{(t)}) \overset{(2.6)}{=} \ell(\theta^{(t)}) + \ell'(\theta^{(t)})(\theta - \theta^{(t)}), \tag{2.7}$$

where the equality holds iff $\theta = \theta^{(t)}$. Therefore, we obtain

$$\begin{aligned}
\ell(\theta) &\overset{(2.4)}{=} h(\theta|\theta^{(t)}) + \int_{\theta^{(t)}}^{\theta} B(z|\theta^{(t)})\,\mathrm{d}z \\
&\overset{(2.7)}{\geqslant} \ell(\theta^{(t)}) + \ell'(\theta^{(t)})(\theta - \theta^{(t)}) + \int_{\theta^{(t)}}^{\theta} B(z|\theta^{(t)})\,\mathrm{d}z \\
&= \ell'(\theta^{(t)})\theta + \int_{\theta^{(t)}}^{\theta} B(z|\theta^{(t)})\,\mathrm{d}z + c^{(t)} \;\triangleq\; Q(\theta|\theta^{(t)}), \qquad (2.8)
\end{aligned}$$

which minorizes $\ell(\theta)$ at $\theta = \theta^{(t)}$, where $c^{(t)}$ is a constant free from θ. The $Q(\theta|\theta^{(t)})$ defined by (2.8) is a desired minorizing function for $\ell(\theta)$.

2.3 Formulation of the SeLF algorithm

By letting $\mathrm{d}Q(\theta|\theta^{(t)})/\,\mathrm{d}\theta = 0$, we can calculate the maximizer of the $Q(\theta|\theta^{(t)})$ function as

$$\begin{aligned}
\theta^{(t+1)} &= \arg\max_{\theta\in\Theta} Q(\theta|\theta^{(t)}) \overset{(2.8)}{=} \arg\max_{\theta\in\Theta} \left\{ \ell'(\theta^{(t)})\theta + \int_{\theta^{(t)}}^{\theta} B(z|\theta^{(t)})\,\mathrm{d}z \right\}, \\
&\overset{(2.3)}{=} \mathrm{sol}\left\{ \ell'(\theta^{(t)}) + \int_{\theta^{(t)}}^{\theta} b(z)\,\mathrm{d}z \;=\; 0,\ \forall\,\theta,\ \theta^{(t)} \in \Theta \right\}, \qquad (2.9)
\end{aligned}$$

which defines a SeLF algorithm. Since the $Q(\theta|\theta^{(t)})$ specified in (2.8) minorizes $\ell(\theta)$ at $\theta = \theta^{(t)}$, by applying the MM principle (Becker *et al.* 1997; Lange *et al.* 2000; Hunter & Lange 2004; Tian *et al.* 2019), we know that the SeLF iteration defined by (2.9) ensuring

$$\ell(\theta^{(t+1)}) \geqslant Q(\theta^{(t+1)}|\theta^{(t)}) \geqslant Q(\theta^{(t)}|\theta^{(t)}) = \ell(\theta^{(t)})$$

for $t = 0, 1, \ldots, \infty$. Hence, the SeLF algorithm is a general principle for iteratively calculating the MLE $\hat{\theta}$ of θ with strongly stable convergence, and its each iteration consists of a second–derivative lower–bound function step (SeLF–step) and a maximization step (M-step), where

SeLF–step: Find a SeLF $b(\theta)$ satisfying (2.2);

M-step: Calculate the maximizer $\theta^{(t+1)}$ of the $Q(\theta|\theta^{(t)})$ function as showed by (2.9).

The two-step process is repeated until convergence occurs.

Theorem 1 below states that the SeLF algorithm holds two characteristics: (i) It strongly stably converges to the MLE $\hat{\theta}$; (ii) It does not depend on any initial values in Θ, in contrast to Newton–Raphson algorithm. Its proof is put in Appendix A.

Theorem 1 (Strongly stable convergence). Let $\hat{\theta}$ be the MLE of θ in the maximization problem (2.1), and $b(\theta)$ be one SeLF for $\ell(\theta)$ in Θ. Then, for any initial value $\theta^{(0)}$ in Θ, the sequence $\{\theta^{(t+1)}\}_{t=0}^{\infty}$ specified by (2.9) strongly stably converges to $\hat{\theta}$. ‖

2.4 Techniques for constructing SeLF functions

We summarize eight log-likelihood function families in Table 1, eight classes Q functions of θ in Table 2 and eight classes equations for the corresponding Q functions in Table 3. For each easy-to-handle part $\ell_0(\theta)$ in Table 1, if we can find a suitable corresponding function $b(\theta)$ to bound the second-order derivatives, then we can obtain an explicit solution for the optimization problem (2.6). Table 2 provides eight corresponding $b(\theta)$ functions to the $\ell_0(\theta)$ parts. From Table 3, we can see that the solutions of Cases 1–6 are equivalent to find a root of quadratic equation.

3. Application 1

3.1 Logarithmic series distribution

The logarithmic series distribution has no probability mass at $x = 0$ and is a limiting distribution of the zero-truncated negative–binomial distribution. Let $\{X_i\}_{i=1}^n \overset{\text{iid}}{\sim} \text{Logseries}(\beta)$ with pmf $[x\log(1+\beta)]^{-1}[\beta/(1+\beta)]^x$, $x = 1, 2, \ldots, \infty$, where $\beta\,(>0)$ is the parameter (Patil 1962). The log-likelihood function is $\ell(\beta) = -\sum_{i=1}^n \log x_i - n\log[\log(1+\beta)] + \sum_{i=1}^n [x_i \log\beta - x_i \log(1+\beta)]$.

First, we define $H_1(\beta) \triangleq \log(1+\beta) - \beta$ for $\beta > 0$. Since $H_1'(\beta) = -\beta/(1+\beta) < 0$, i.e., $H_1(\beta)$ is a strictly decreasing function for $\beta > 0$, we have

$$\log(1+\beta) - \beta = H_1(\beta) < H_1(0) = 0 \quad \text{or} \quad \frac{1}{\log(1+\beta)} > \frac{1}{\beta} > 0. \tag{3.1}$$

Next, note that $z + z^2$ is strictly increasing for $z > 0$, then we have

$$\frac{1}{\log(1+\beta)} + \frac{1}{\log^2(1+\beta)} \overset{(3.1)}{>} \frac{1}{\beta} + \frac{1}{\beta^2}. \tag{3.2}$$

The first– and second–order derivatives of $\ell(\beta)$ are given by

$$\begin{aligned}
\ell'(\beta) &= -\frac{n}{(1+\beta)\log(1+\beta)} + \frac{n\bar{x}}{\beta} - \frac{n\bar{x}}{1+\beta} \quad \text{and} \\
\ell''(\beta) &= \frac{n}{(1+\beta)^2}\left[\frac{1}{\log(1+\beta)} + \frac{1}{\log^2(1+\beta)}\right] - \frac{n\bar{x}}{\beta^2} + \frac{n\bar{x}}{(1+\beta)^2} \\
&\overset{(3.2)}{>} \frac{n}{(1+\beta)^2}\left(\frac{1}{\beta} + \frac{1}{\beta^2}\right) - \frac{n\bar{x}}{\beta^2} + \frac{n\bar{x}}{(1+\beta)^2} = \frac{n}{(1+\beta)\beta^2} - \frac{n\bar{x}}{\beta^2} + \frac{n\bar{x}}{(1+\beta)^2} \\
&= \frac{n}{2}\left[\frac{1}{(1+\beta)\beta^2} + \frac{1}{(1+\beta)\beta^2}\right] - \frac{n\bar{x}}{\beta^2} + \frac{n\bar{x}}{(1+\beta)^2} \\
&> \frac{n}{2}\left[\frac{1}{(1+\beta)\beta^2} + \frac{1}{(1+\beta)^2\beta}\right] - \frac{n\bar{x}}{\beta^2} + \frac{n\bar{x}}{(1+\beta)^2} \\
&= \frac{n}{2}\frac{(\beta+1)^2 - \beta^2}{(1+\beta)^2\beta^2} - \frac{n\bar{x}}{\beta^2} + \frac{n\bar{x}}{(1+\beta)^2} = \frac{n}{2}\frac{1}{\beta^2} - \frac{n}{2}\frac{1}{(1+\beta)^2} - \frac{n\bar{x}}{\beta^2} + \frac{n\bar{x}}{(1+\beta)^2} \\
&= \left(\frac{n}{2} - n\bar{x}\right)\left[\frac{1}{\beta^2} - \frac{1}{(1+\beta)^2}\right] = \frac{a_1}{\beta^2} - \frac{a_1}{(1+\beta)^2},
\end{aligned}$$

where $a_1 \triangleq n(0.5 - \bar{x})$. Therefore, the SeLF iteration is

$$\begin{aligned}
\beta^{(t+1)} &\overset{(2.9)}{=} \text{sol}\left\{\ell'(\beta^{(t)}) - \frac{a_1}{\beta(1+\beta)} + \frac{a_1}{\beta^{(t)}(1+\beta^{(t)})} = 0,\ \forall\, \beta, \beta^{(t)} > 0\right\} \\
&= \text{sol}\left\{a_2^{(t)}\beta^2 + a_2^{(t)}\beta - a_1 = 0,\ \forall\, \beta, \beta^{(t)} > 0\right\},
\end{aligned}$$

where

$$a_2^{(t)} \triangleq \ell'(\beta^{(t)}) + \frac{a_1}{\beta^{(t)}(1+\beta^{(t)})}.$$

3.2 Unit power-logarithmic distribution

Let $\{X_i\}_{i=1}^n \overset{\text{iid}}{\sim} \text{UPL}(\alpha)$ with pdf $\log^{-1}(\alpha+1)\cdot(1-x^\alpha)[-\log(x)]^{-1}$, $x \in (0,1)$, where $\alpha\,(>0)$ is the parameter. The log-likelihood function of α is $\ell(\alpha) = -n\log[\log(\alpha+1)] + \sum_{i=1}^n \log(1-x_i^\alpha) - \sum_{i=1}^n \log[-\log(x_i)]$.

First, we define $H_2(x) \triangleq x/(1-x)^2$ for $x \in (0,1)$. Since $H_2'(x) = (1+x)/(1-x)^3 > 0$, i.e., $H_2(x)$ is strictly increasing for $x \in (0,1)$, we have

$$\frac{x_i^\alpha}{(1-x_i^\alpha)^2} = H_2(x_i^\alpha) \leqslant H_2(x_{(n)}^\alpha) = \frac{x_{(n)}^\alpha}{(1-x_{(n)}^\alpha)^2}, \tag{3.3}$$

where $x_{(n)} \triangleq \max(x_1, \ldots, x_n)$. The first– and second–order derivatives of $\ell(\alpha)$ are given by

$$\begin{aligned}
\ell'(\alpha) &= -\frac{n}{(\alpha+1)\log(\alpha+1)} - \sum_{i=1}^n \frac{x_i^\alpha \log x_i}{1-x_i^\alpha} \quad \text{and} \\
\ell''(\alpha) &= n\frac{\log(\alpha+1)+1}{(\alpha+1)^2\log^2(\alpha+1)} - \sum_{i=1}^n \frac{x_i^\alpha}{(1-x_i^\alpha)^2}\log^2 x_i \\
&\overset{(3.3)}{\geqslant} 0 - \frac{x_{(n)}^\alpha \log x_{(n)}}{(1-x_{(n)}^\alpha)^2} \cdot \frac{\sum_{i=1}^n \log^2 x_i}{\log x_{(n)}},
\end{aligned}$$

respectively. Therefore, the SeLF iteration is

$$\alpha^{(t+1)} = \text{sol}\left\{\ell'(\alpha^{(t)}) + \left(\frac{1}{1-x_{(n)}^\alpha} - \frac{1}{1-x_{(n)}^{\alpha^{(t)}}}\right)\frac{\sum_{i=1}^n \log^2 x_i}{\log x_{(n)}} = 0,\ \forall\, \alpha, \alpha^{(t)} > 0\right\}, \tag{3.4}$$

which has an explicit solution.

3.3 Burr Type XII distribution

As a mixture of the Weibull distribution and a standard gamma distribution, the Burr Type XII distribution often exhibits a good fit when it is used to model biological, clinical, reliability or other experimental data (Burr 1942; Wingo 1983; Wingo 1993). Let $\{X_i\}_{i=1}^n \overset{\text{iid}}{\sim} \text{Burr}^{(\text{XII})}(\alpha, \beta)$ with pdf $\alpha\beta x^{\alpha-1}/(x^\alpha+1)^{\beta+1}$ for $x > 0$, where $\alpha > 0$ and $\beta > 0$ are two unknown parameters. The log-likelihood function is $\ell(\alpha, \beta) = n\log(\alpha) + n\log(\beta) + n(\alpha - 1)G_1(\boldsymbol{x}) - (\beta+1)\sum_{i=1}^n \log(x_i^\alpha + 1)$, where $G_1(\boldsymbol{x}) \triangleq (1/n)\sum_{i=1}^n \log x_i$. Given α, we have $\hat{\beta} = n/\sum_{i=1}^n \log(x_i^\alpha + 1)$. Given β, the conditional log-likelihood function of α is

$$\ell(\alpha) \triangleq \ell(\alpha|\beta) = n\log(\alpha) + n\alpha G_1(\boldsymbol{x}) - (\beta+1)\sum_{i=1}^n \log(x_i^\alpha + 1) + \text{constant},$$

and its first– and second–order derivatives are

$$\begin{aligned}
\ell'(\alpha) &= \frac{n}{\alpha} + nG_1(\boldsymbol{x}) - (\beta+1)\sum_{i=1}^{n}\frac{x_i^\alpha \log x_i}{x_i^\alpha + 1} \quad \text{and} \\
\ell''(\alpha) &= -\frac{n}{\alpha^2} - (\beta+1)\sum_{i=1}^{n}\frac{x_i^\alpha \log^2 x_i}{(x_i^\alpha+1)^2} \;=\; -\frac{n}{\alpha^2} - (\beta+1)\sum_{i=1}^{n}\frac{\log^2 x_i}{\left(x_i^{\alpha/2} + x_i^{-\alpha/2}\right)^2} \\
&\geqslant -\frac{n}{\alpha^2} - \frac{\beta+1}{4}\sum_{i=1}^{n}\log^2 x_i \;=\; -\frac{n}{\alpha^2} - a_3, \qquad (3.5)
\end{aligned}$$

where $a_3 \triangleq n(\beta+1)G_2(\boldsymbol{x})/4$ and $G_2(\boldsymbol{x}) \triangleq (1/n)\sum_{i=1}^{n}\log^2 x_i$ and the last inequality holds because $x_i^{\alpha/2} + x_i^{-\alpha/2} \geqslant 2$ for all $\alpha > 0$. Therefore, the SeLF iteration for α is

$$\alpha^{(t+1)} \;=\; \mathrm{sol}\left\{a_3\alpha^2 - a_4^{(t)}\alpha - n = 0,\ \forall\, \alpha, \alpha^{(t)} > 0\right\}.$$

where $a_4^{(t)} \triangleq a_3\alpha^{(t)} - n/\alpha^{(t)} + \ell'(\alpha^{(t)})$.

3.4 Gumbel–maximum distribution

Let $\{X_i\}_{i=1}^n \overset{\text{iid}}{\sim}$ Gumbel–Max(μ,σ) with pdf $\sigma^{-1}\mathrm{e}^{-(x-\mu)/\sigma}\exp\left[-\mathrm{e}^{-(x-\mu)/\sigma}\right]$ for $x \in \mathbb{R}$, where $\mu \in \mathbb{R}$ is the location parameter and $\sigma > 0$ is the scale parameter. The log-likelihood function is $\ell(\mu,\sigma) = -n\log\sigma - \sum_{i=1}^n (x_i-\mu)/\sigma - \sum_{i=1}^n \exp\left[-(x_i-\mu)/\sigma\right]$. Given σ, we have $\hat{\mu} = -\sigma\log\left(\sum_{i=1}^n \mathrm{e}^{-x_i/\sigma}/n\right)$. Given μ and set $\beta = 1/\sigma$, we have

$$\ell(\beta) \;\triangleq\; \ell(\beta|\mu) = n\log\beta - \sum_{i=1}^{n} z_i\beta - \sum_{i=1}^{n}\mathrm{e}^{-z_i\beta},$$

where $z_i \triangleq x_i - \mu$ for $i = 1,\ldots,n$. Define $\mathbb{I}_0 \triangleq \{i\colon\ z_i \leqslant 0,\ i = 1,\ldots,n\}$, $\mathbb{I}_1 \triangleq \{i\colon\ z_i > 0,\ i = 1,\ldots,n\}$ and $n_1 \triangleq \sum_{i=1}^n I(z_i > 0) = \#\mathbb{I}_1$. The first– and second–order derivatives of $\ell(\beta)$ are

$$\begin{aligned}
\ell'(\beta) &= \frac{n}{\beta} - \sum_{i=1}^{n} z_i + \sum_{i=1}^{n} z_i\mathrm{e}^{-z_i\beta}, \\
\ell''(\beta) &= -\frac{n}{\beta^2} - \sum_{i\in\mathbb{I}_0}^{n} z_i^2\mathrm{e}^{-z_i\beta} - \sum_{i\in\mathbb{I}_1}^{n} z_i^2\mathrm{e}^{-z_i\beta}
\end{aligned}$$

$$
\begin{aligned}
&\overset{\text{(B.5) \& (B.7)}}{\geqslant} && -\frac{n}{\beta^2} - \sum_{i\in\mathbb{I}_0} z_i^2 \frac{4\mathrm{e}^{2\max(-z_i\beta^{(t)}-1,\,0)}}{z_i^2(2\beta^{(t)}-\beta)^2} - \sum_{i\in\mathbb{I}_1} z_i^2 \frac{2}{3z_i^2\beta^2} \\
&= && -\frac{1}{\beta^2}\left(n + \frac{2n_1}{3}\right) - \frac{1}{(2\beta^{(t)}-\beta)^2} \sum_{i\in\mathbb{I}_0} 4\mathrm{e}^{2\max(-z_i\beta^{(t)}-1,\,0)} \\
&= && -\frac{a_5}{\beta^2} - \frac{a_6^{(t)}}{(2\beta^{(t)}-\beta)^2},
\end{aligned}
$$

where $a_5 \triangleq n + 2n_1/3$ and $a_6^{(t)} \triangleq \sum_{i\in\mathbb{I}_0} 4\mathrm{e}^{2\max(-z_i\beta^{(t)}-1,\,0)}$. Thus, the SeLF iteration is equivalent to finding the root of the following quadratic equation:

$$
1/\sigma^{(t+1)} = \beta^{(t+1)} = \mathrm{sol}\left\{a_7^{(t)}\beta^2 - a_8^{(t)}\beta - 2a_5\beta^{(t)} = 0,\ \forall\,\beta, \beta^{(t)} > 0\right\}, \tag{3.6}
$$

where $a_7^{(t)} \triangleq (a_6^{(t)} - a_5)/\beta^{(t)} + \ell'(\beta^{(t)})$ and $a_8^{(t)} \triangleq 2\beta^{(t)}\ell'(\beta^{(t)}) - 3a_5 + a_6^{(t)}$.

3.5 Weibull distribution

Let $\{X_i\}_{i=1}^n \overset{\text{iid}}{\sim} \mathrm{Weibull}(\theta, \lambda)$ with pdf $\theta x^{\theta-1}\lambda^{-\theta}\exp[-(x/\lambda)^\theta]$ for $x > 0$, where $\theta\,(> 0)$ and $\lambda\,(> 0)$ are two unknown parameters. The log-likelihood function is $\ell(\theta, \lambda) = n\log(\theta) + (\theta - 1)\sum_{i=1}^n \log(x_i) - n\theta\log(\lambda) - \sum_{i=1}^n (x_i/\lambda)^\theta$. Given θ, we have $\lambda^\theta = \sum_{i=1}^n x_i^\theta/n$. Given λ, we have

$$
\begin{aligned}
\ell(\theta) \ \triangleq\ \ell(\theta|\lambda) \ &= \ n\log(\theta) + \theta\sum_{i=1}^n \log\left(\frac{x_i}{\lambda}\right) - \sum_{i=1}^n \left(\frac{x_i}{\lambda}\right)^\theta - \sum_{i=1}^n \log(x_i) \\
&= \ n\log(\theta) + \theta\sum_{i=1}^n z_i - \sum_{i=1}^n \exp(z_i\theta) + \text{constant},
\end{aligned}
$$

where $z_i \triangleq \log(x_i/\lambda)$ for $i = 1, \ldots, n$. Define $\mathbb{I}_0 \triangleq \{i\colon 0 < x_i/\lambda \leqslant 1,\ i = 1, \ldots, n\}$, $\mathbb{I}_1 \triangleq \{i\colon x_i/\lambda > 1,\ i = 1, \ldots, n\}$ and $n_0 \triangleq \sum_{i=1}^n I(0 < x_i/\lambda \leqslant 1) = \#\mathbb{I}_0$. The first– and second–order derivatives of $\ell(\theta)$ are given by

$$
\begin{aligned}
\ell'(\theta) \ &= \ \frac{n}{\theta} + \sum_{i=1}^n z_i - \sum_{i=1}^n z_i\exp(z_i\theta) \quad \text{and} \\
\ell''(\theta) \ &= \ -\frac{n}{\theta^2} - \sum_{i=1}^n z_i^2\exp(z_i\theta) \ = \ -\frac{n}{\theta^2} - \sum_{i\in\mathbb{I}_0} z_i^2\exp(z_i\theta) - \sum_{i\in\mathbb{I}_1} z_i^2\exp(z_i\theta)
\end{aligned}
$$

$$
\begin{aligned}
&\overset{\text{(B.7) \& (B.5)}}{\geqslant} && -\frac{n}{\theta^2} - \sum_{i\in\mathbb{I}_0} z_i^2 \frac{2}{3z_i^2\theta^2} - \sum_{i\in\mathbb{I}_1} z_i^2 \frac{4\mathrm{e}^{2\max(z_i\theta^{(t)}-1,\,0)}}{(2\theta^{(t)}-\theta)^2 z_i^2} \\
&= && -\frac{1}{\theta^2}\left(n + \frac{2n_0}{3}\right) - \frac{1}{(2\theta^{(t)}-\theta)^2} \sum_{i\in\mathbb{I}_1} 4\mathrm{e}^{2\max(z_i\theta^{(t)}-1,\,0)} \\
&= && -\frac{a_9}{\theta^2} - \frac{a_{10}^{(t)}}{(2\theta^{(t)}-\theta)^2},
\end{aligned}
$$

where $a_9 \triangleq n + 2n_0/3$ and $a_{10}^{(t)} \triangleq \sum_{i\in\mathbb{I}_1} 4\exp[2\max(z_i\theta^{(t)}-1,\,0)]$. So, the SeLF iteration is

$$
\theta^{(t+1)} \;=\; \mathrm{sol}\left\{a_{11}^{(t)}\theta^2 - a_{12}^{(t)}\theta - 2a_9\theta^{(t)} = 0,\ \forall\,\theta, \theta^{(t)} > 0\right\}.
$$

where $a_{11}^{(t)} \triangleq (a_{10}^{(t)} - a_9)/\theta^{(t)} + \ell'(\theta^{(t)})$ and $a_{12}^{(t)} \triangleq 2\theta^{(t)}\ell'(\theta^{(t)}) - 3a_9 + a_{10}^{(t)}$.

4. Three accelerating SeLF algorithms

4.1 Optimal SeLF algorithm

From the definition of a SeLF in (2.2), it is possible that there exist many (even infinite) SeLFs for $\ell(\theta)$, e.g., if $\ell''(\theta) \geqslant b(\theta) \geqslant 0$ for all $\theta \in \Theta$, then for any $c \in (0,1)$, the function $c\,b(\theta)$ is also a SeLF for $\ell(\theta)$. Next, Theorem 3 in Section 6 shows that the convergence rates of the SeLF algorithm is linear relying on the ratio $\ell''(\hat\theta)/b(\hat\theta)$. More specifically, when $\ell''(\hat\theta)/b(\hat\theta)$ tends to 1, the convergence rate will be super-linear.

To develop an accelerating SeLF algorithm, we can borrow the idea of constructing a minorizing Q-function in MM algorithms. In other words, we can construct a SeLF $b(\cdot|\theta^{(t)})$ by satisfying $\ell''(\theta^{(t)}) = b(\theta^{(t)}|\theta^{(t)})$ for all $\theta^{(t)} \in \Theta$.

Definition 1 (Optimal SeLF of $\ell(\theta)$ at $\theta = \theta^{(t)}$). Let $\theta^{(t)}$ denote the t-iteration of the MLE $\hat\theta \triangleq \arg\max_{\theta\in\Theta} \ell(\theta)$ and the function $b(\theta|\theta^{(t)})$ be a SeLF of the log-likelihood $\ell(\theta)$, i.e.,

$$
\ell''(\theta) \geqslant b(\theta|\theta^{(t)}), \quad \forall\,\theta, \theta^{(t)} \in \Theta. \tag{4.1}
$$

If the $b(\theta|\theta^{(t)})$ function further satisfies

$$
\ell''(\theta^{(t)}) = b(\theta^{(t)}|\theta^{(t)}), \tag{4.2}
$$

We call $b(\theta|\theta^{(t)})$ an optimal SeLF of $\ell(\theta)$ at $\theta = \theta^{(t)}$. $\|$

From (4.1) and (4.2), we know that the $b(\theta|\theta^{(t)})$ is a minorizing function of $\ell''(\theta)$ at $\theta = \theta^{(t)}$. We say a SeLF iteration generated by using an optimal SeLF is an optimal SeLF iteration and the corresponding algorithm is an *optimal* SeLF algorithm. Theorem 4 in Section 6 states that the optimal SeLF algorithm possesses a quadratic convergence rate. To illustrate the usefulness of the optimal SeLF algorithm, we will provide two examples, Yule–Simon and Birnbaum–Saunders distributions, in Section 5.

4.2 Sub-optimal SeLF algorithm

In some situations, we can hardly find the optimal SeLF for $\ell(\theta)$. Then, we can first relax $\ell''(\theta)$ such that $\ell''(\theta) \geqslant \tilde{\ell}''(\theta)$, and then construct a minorizing function $b(\theta|\theta^{(t)})$ for the relaxed $\tilde{\ell}''(\theta)$ at $\theta = \theta^{(t)}$. Since $\tilde{\ell}''(\theta)$ majorizes the lower bound function $b(\theta|\theta^{(t)})$, which is called a *sub-optimal* SeLF for the $\ell(\theta)$. The SeLF iteration/algorithm obtained by using a sub-optimal SeLF is named a *sub-optimal* SeLF iteration/algorithm. The sub-optimal SeLF algorithm has a linear convergence rate because $\ell''(\hat{\theta})/b(\hat{\theta}|\hat{\theta}) < 1$, while the speed of convergence is improved due to the growth of $\ell''(\hat{\theta})/b(\hat{\theta}|\hat{\theta})$. To illustrate the usefulness of the sub-optimal SeLF algorithm, we provide two examples, Dirichlet and inverse gamma distributions, in Section 5.

4.3 A fast–SeLF algorithm with weakly stable convergence

Theorems 3 in Section 6 stated that the SeLF algorithm holds a linear convergence rate only. To accelerate its convergence speed, we propose an acceleration technique with weakly stable convergence.

Theorem 2 (The fast–SeLF algorithm with weakly stable convergence). Let $\hat{\theta}$ be the MLE of θ by maximizing the log-likelihood function $\ell(\theta)$, and the Q-function $Q(\theta|\theta^{(t)})$ be specified by (2.8). The fast SeLF iteration is defined as

$$\theta^{(t+1)} = \theta^{(t)} + s(\theta^{(t)})(\tilde{\theta}^{(t+1)} - \theta^{(t)}) = [1 - s(\theta^{(t)})]\theta^{(t)} + s(\theta^{(t)})\tilde{\theta}^{(t+1)}, \tag{4.3}$$

where $s(\theta^{(t)})$ is said to be the step length, $\tilde{\theta}^{(t+1)} - \theta^{(t)}$ is called a direction,

$$s(\theta^{(t)}) \triangleq \min\left\{\frac{b(\theta^{(t)})}{\ell''(\theta^{(t)})} I(\ell''(\theta^{(t)}) < 0) + I(\ell''(\theta^{(t)}) \geqslant 0),\ 2\right\} \quad \text{and} \tag{4.4}$$

$$\tilde{\theta}^{(t+1)} \triangleq \operatorname{sol}\left\{\ell'(\theta^{(t)}) + \int_{\theta^{(t)}}^{\theta} b(z)\,\mathrm{d}z \ = \ 0,\ \forall\,\theta,\ \theta^{(t)} \in \Theta\right\}. \tag{4.5}$$

Thus,

(i) For any initial value $\theta^{(0)}$ in Θ, the sequence $\{\theta^{(t+1)}\}_{t=0}^{\infty}$ specified by (4.3) weakly converges to $\hat{\theta}$. In other words, the fast–SeLF algorithm does not depend on any initial values in Θ.

(ii) If $b(\hat{\theta})/\ell''(\hat{\theta}) > 2$, then the convergence speed $(2\ell''(\hat{\theta})/b(\hat{\theta}))$ of the fast–SeLF algorithm is twice of the convergence speed $(\ell''(\hat{\theta})/b(\hat{\theta}))$ of the SeLF algorithm. If $1 \leqslant b(\hat{\theta})/\ell''(\hat{\theta}) \leqslant 2$, then the fast–SeLF algorithm has a super–linear convergence rate comparing with a linear convergence rate for the SeLF algorithm. ‖

Remark 1 (A comment on Theorem 2). Obviously, for the optimal SeLF algorithm, because of (4.2), we have $b(\theta^{(t)}|\theta^{(t)})/\ell''(\theta^{(t)}) = 1$, so that $\theta^{(t+1)} = \tilde{\theta}^{(t+1)}$. Therefore, our acceleration technique is useless for the optimal SeLF algorithm. ‖

5. Application 2

5.1 Yule–Simon distribution

The Yule–Simon distribution (Yule 1925; Garcia 2011) can be used to model species among genera, words frequencies in texts, numbers of papers published by researchers, cities by population, income by size (Leisen *et al.* 2017). Let $\{X_i\}_{i=1}^{n} \overset{\text{iid}}{\sim} \mathrm{YS}(\theta)$ with pmf $\theta \cdot B(x, \theta+1)$, $x = 1, 2, \ldots, \infty$, where $\theta\,(> 0)$ is the shape parameter. The log-likelihood function is $\ell(\theta) = n\log(\theta) + n\log\Gamma(\theta+1) - \sum_{i=1}^{n}\log\Gamma(x_i+\theta+1) + \text{constant}$.

Let $\psi(\cdot)$ denote the digamma function defined as

$$\psi(\alpha) \triangleq \frac{\Gamma'(\alpha)}{\Gamma(\alpha)} = -\gamma + \sum_{m=0}^{\infty}\left(\frac{1}{m+1} - \frac{1}{m+\alpha}\right), \tag{5.1}$$

where $\gamma = \lim_{n\to\infty}\left(\sum_{m=1}^{n} m^{-1} - \log n\right) \approx 0.5772$ is the Euler–Mascheroni constant. The first– and second–order derivatives of $\ell(\theta)$ are

$$\begin{aligned}
\ell'(\theta) &= \frac{n}{\theta} + \sum_{i=1}^{n}[\psi(\theta+1) - \psi(x_i+\theta+1)] \\
&\overset{(5.1)}{=} \frac{n}{\theta} + \sum_{i=1}^{n}\left(-\sum_{m=0}^{\infty}\frac{1}{m+\theta+1} + \sum_{m=0}^{\infty}\frac{1}{m+x_i+\theta+1}\right) \qquad [\text{Let } m+x_i=k] \\
&= \frac{n}{\theta} + \sum_{i=1}^{n}\left(-\sum_{m=0}^{\infty}\frac{1}{m+\theta+1} + \sum_{k=x_i}^{\infty}\frac{1}{k+\theta+1}\right) \\
&= \frac{n}{\theta} + \sum_{i=1}^{n}\left(-\sum_{m=0}^{x_i-1}\frac{1}{m+\theta+1}\right), \\
\ell''(\theta) &= -\frac{n}{\theta^2} + \sum_{i=1}^{n}\sum_{m=1}^{x_i}(m+\theta)^{-2} \overset{(B.4)}{\geqslant} -\frac{n}{\theta^2} + a_{13}^{(t)}\left(a_{14}^{(t)}+\theta\right)^{-2} \triangleq b(\theta|\theta^{(t)}),
\end{aligned}$$

where the equality holds iff $\theta = \theta^{(t)}$, implying that $b(\theta|\theta^{(t)})$ is an optimal SeLF of $\ell(\theta)$ at $\theta = \theta^{(t)}$, and here

$$s_{im}^{(t)} \triangleq \frac{1}{m+\theta^{(t)}}, \qquad S^{(t)} \triangleq \sum_{i=1}^{n}\sum_{m=1}^{x_i}(m+\theta^{(t)})^{-2},$$

$$a_{13}^{(t)} \triangleq \frac{S^{(t)3}}{\left(\sum_{i=1}^{n}\sum_{m=1}^{x_i} s_{im}^{(t)3}\right)^2} \quad \text{and} \quad a_{14}^{(t)} \triangleq \frac{\sum_{i=1}^{n}\sum_{m=1}^{x_i} m s_{im}^{(t)3}}{\sum_{i=1}^{n}\sum_{m=1}^{x_i} s_{im}^{(t)3}}.$$

Therefore, the iteration for calculating the MLE $\hat{\theta}$ is

$$\begin{aligned}
\theta^{(t+1)} &= \text{sol}\left\{\ell'(\theta^{(t)}) + \int_{\theta^{(t)}}^{\theta} b(z|\theta^{(t)})\,\mathrm{d}z = 0,\ \forall\,\theta,\theta^{(t)} > 0\right\} \\
&= \text{sol}\left\{\ell'(\theta^{(t)}) + \frac{n}{\theta} - \frac{a_{13}^{(t)}}{a_{14}^{(t)}+\theta} - \frac{n}{\theta^{(t)}} + \frac{a_{13}^{(t)}}{a_{14}^{(t)}+\theta^{(t)}} = 0,\ \forall\,\theta,\theta^{(t)} > 0\right\} \\
&= \text{sol}\left\{a_{15}^{(t)}\theta^2 + (n - a_{13}^{(t)} + a_{14}^{(t)}a_{15}^{(t)})\theta + n a_{14}^{(t)} = 0,\ \forall\,\theta,\theta^{(t)} > 0\right\}, \qquad (5.2)
\end{aligned}$$

where $a_{15}^{(t)} \triangleq \ell'(\theta^{(t)}) - n/\theta^{(t)} + a_{13}^{(t)}/(a_{14}^{(t)}+\theta^{(t)})$.

5.2 Birnbaum–Saunders distribution

The *Birnbaum–Saunders* (BS) distribution plays an important role in lifetime data analysis, environmental science and biological models (Engelhardt *et al.* 1981; Leiva 2015; Balakrishnan & Kundu 2019). Let $\{X_i\}_{i=1}^n \overset{\text{iid}}{\sim} \text{BS}(\alpha, \beta)$ with pdf $(2\sqrt{2\pi}\alpha\beta)^{-1}[(\beta/x)^{1/2} + (\beta/x)^{3/2}]\exp[-(x/\beta + \beta/x - 2)/(2\alpha^2)]$ for $x > 0$, where $\alpha\,(> 0)$ and $\beta\,(> 0)$ are two parameters. The log-likelihood function is

$$\ell(\alpha, \beta) = -n\log\alpha - n\log\beta + \sum_{i=1}^n \log[(\beta/x_i)^{1/2} + (\beta/x_i)^{3/2}] - \frac{1}{2\alpha^2}\sum_{i=1}^n (x_i/\beta + \beta/x_i - 2).$$

Given β, we have $\alpha^2 = (1/n)\sum_{i=1}^n (x_i/\beta + \beta/x_i - 2)$. Given α, we have

$$\ell(\beta) \;\triangleq\; \ell(\beta|\alpha) = -\frac{n}{2}\log\beta + \sum_{i=1}^n \log\left(1 + \frac{\beta}{x_i}\right) - \frac{1}{2\alpha^2}\sum_{i=1}^n \left(\frac{x_i}{\beta} + \frac{\beta}{x_i}\right) + \text{constant}.$$

The first– and second–order derivatives of $\ell(\beta)$ are

$$\begin{aligned}
\ell'(\beta) &= -\frac{n}{2\beta} + \sum_{i=1}^n \frac{1}{x_i + \beta} + \frac{1}{2\alpha^2}\sum_{i=1}^n \left(\frac{x_i}{\beta^2} - \frac{1}{x_i}\right) \quad \text{and} \\
\ell''(\beta) &= \frac{n}{2\beta^2} - \sum_{i=1}^n \frac{1}{(x_i + \beta)^2} - \frac{n\bar{x}}{\alpha^2\beta^3} \\
&\overset{\text{(B.3)}}{\geqslant} \frac{n}{2\beta^2} - \sum_{i=1}^n \left[\frac{(1 - r_i^{(t)})^3}{x_i^2} + \frac{r_i^{(t)3}}{\beta^2}\right] - \frac{n\bar{x}}{\alpha^2\beta^3} \\
&= \left(\frac{n}{2} - \sum_{i=1}^n r_i^{(t)3}\right)\frac{1}{\beta^2} - \left(\frac{n\bar{x}}{\alpha^2}\right)\frac{1}{\beta^3} - \sum_{i=1}^n \frac{(1 - r_i^{(t)})^3}{x_i^2} \\
&= \frac{a_{16}^{(t)}}{\beta^2} - \frac{a_{17}}{\beta^3} - a_{18}^{(t)} \;\triangleq\; b(\beta|\beta^{(t)}),
\end{aligned}$$

where the equality holds iff $\beta = \beta^{(t)}$, implying that $b(\beta|\beta^{(t)})$ is an optimal SeLF of $\ell(\beta)$ at $\beta = \beta^{(t)}$, and here

$$r_i^{(t)} \triangleq \frac{\beta^{(t)}}{x_i + \beta^{(t)}}, \quad a_{16}^{(t)} \triangleq \frac{n}{2} - \sum_{i=1}^n r_i^{(t)3}, \quad a_{17} \triangleq \frac{n\bar{x}}{\alpha^2} \quad \text{and} \quad a_{18}^{(t)} \triangleq \sum_{i=1}^n \frac{(1 - r_i^{(t)})^3}{x_i^{-2}}.$$

Thus, the SeLF iteration is

$$\beta^{(t+1)} = \text{sol}\left\{-a_{18}^{(t)}\beta^3 + a_{19}^{(t)}\beta^2 - a_{16}^{(t)}\beta + a_{17}/2 = 0,\ \forall\,\beta, \beta^{(t)} > 0\right\},$$

where $a_{19}^{(t)} \triangleq \ell'(\beta^{(t)}) + a_{16}^{(t)}/\beta^{(t)} - a_{17}/(2\beta^{(t)2}) + a_{18}^{(t)}\beta^{(t)}$.

5.3 Dirichlet distribution

Let $\mathbf{x}_i = (X_{i1}, \ldots, X_{id})^\top$ and $\{\mathbf{x}_i\}_{i=1}^n \overset{\text{iid}}{\sim} \text{Dirichlet}_d(\boldsymbol{\alpha})$ with joint pdf

$$\frac{\Gamma(\alpha_+)}{\prod_{j=1}^d \Gamma(\alpha_j)} \prod_{j=1}^d x_{ij}^{\alpha_j - 1}, \quad \boldsymbol{x}_i = (x_{i1}, \ldots, x_{id})^\top \in \mathbb{T}_d,$$

where $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_d)^\top$ is the parameter vector, each $\alpha_j > 0$ and $\alpha_+ \triangleq \sum_{j=1}^d \alpha_j$. The log-likelihood function is $\ell(\boldsymbol{\alpha}) = n[\log\Gamma(\alpha_+) - \sum_{j=1}^d \log\Gamma(\alpha_j) + \sum_{j=1}^d (\alpha_j - 1)\log(G_j)]$, where $G_j = (\prod_{i=1}^n x_{ij})^{1/n}$ for $j = 1, \ldots, d$. Given $\{\alpha_1, \ldots, \alpha_{j-1}, \alpha_{j+1}, \ldots, \alpha_d\}$, to calculate the MLE of α_j, we define $\ell_j(\alpha_j) \triangleq \ell(\alpha_j | \alpha_1, \ldots, \alpha_{j-1}, \alpha_{j+1}, \ldots, \alpha_d)$ and then we have

$$\begin{aligned}
\ell_j'(\alpha_j) &= n\big[\psi(\alpha_+) - \psi(\alpha_j) + \log(G_j)\big] \\
&\overset{(5.1)}{=} -n\sum_{m=0}^{\infty}(m+\alpha_+)^{-1} + n\sum_{m=0}^{\infty}(m+\alpha_j)^{-1} + n\log(G_j) \quad \text{and} \\
\ell_j''(\alpha_j) &= n\sum_{m=0}^{\infty}(m+\alpha_+)^{-2} - n\sum_{m=0}^{\infty}(m+\alpha_j)^{-2} \\
&= n\sum_{m=0}^{\infty}(m+\alpha_j+\alpha_{-j})^{-2} - n\sum_{m=0}^{\infty}(m+\alpha_j)^{-2} \\
&\geqslant n\sum_{m=0}^{\infty}(m+\alpha_j+\lceil\alpha_{-j}\rceil)^{-2} - n\sum_{m=0}^{\infty}(m+\alpha_j)^{-2} \\
&= -n\sum_{m=0}^{\lceil\alpha_{-j}\rceil-1}(m+\alpha_j)^{-2} = -\frac{n}{\alpha_j^2} - n\sum_{m=1}^{\lceil\alpha_{-j}\rceil-1}(m+\alpha_j)^{-2} \\
&\overset{(B.3)}{\geqslant} -\frac{n}{\alpha_j^2} - \frac{n\sum_{m=1}^{\lceil\alpha_{-j}\rceil-1} r_{jm}^{(t)3}}{\alpha_j^2} - n\sum_{m=1}^{\lceil\alpha_{-j}\rceil-1}\frac{(1-r_{jm}^{(t)})^3}{m^2} \\
&= \frac{a_{20,j}^{(t)}}{\alpha_j^2} + a_{21,j}^{(t)} \triangleq b(\alpha_j | \alpha_j^{(t)}),
\end{aligned}$$

where $\alpha_{-j} \triangleq \alpha_+ - \alpha_j$, $\lceil w \rceil$ denotes the least integer greater than or equal to w,

$$r_{jm}^{(t)} \triangleq \frac{\alpha_j^{(t)}}{m + \alpha_j^{(t)}}, \quad a_{20,j}^{(t)} \triangleq -n - n\sum_{m=1}^{\lceil\alpha_{-j}\rceil-1} r_{jm}^{(t)3}, \quad a_{21,j}^{(t)} \triangleq -n\sum_{m=1}^{\lceil\alpha_{-j}\rceil-1}\frac{(1-r_{jm}^{(t)})^3}{m^{-2}}.$$

Thus, the sub-optimal SeLF iteration is

$$\begin{aligned}
\alpha_j^{(t+1)} &= \operatorname{sol}\left\{\ell_j'(\alpha_j^{(t)}) + \int_{\alpha_j^{(t)}}^{\alpha_j} b(z|\alpha_j^{(t)})\,\mathrm{d}z = 0,\ \forall\,\alpha_j, \alpha_j^{(t)} > 0\right\} \\
&= \operatorname{sol}\left\{\ell_j'(\alpha_j^{(t)}) - \frac{a_{20,j}^{(t)}}{\alpha_j} + \frac{a_{20,j}^{(t)}}{\alpha_j^{(t)}} + a_{21,j}^{(t)}(\alpha_j - \alpha_j^{(t)}) = 0,\ \forall\,\alpha_j, \alpha_j^{(t)} > 0\right\} \\
&= \operatorname{sol}\left\{a_{21,j}^{(t)}\alpha_j^2 + a_{22,j}^{(t)}\alpha_j - a_{20,j}^{(t)} = 0,\ \forall\,\alpha_j, \alpha_j^{(t)} > 0\right\},
\end{aligned}$$

where $a_{22,j}^{(t)} \triangleq \ell_j'(\alpha_j^{(t)}) + a_{20,j}^{(t)}/\alpha_j^{(t)} - a_{21,j}^{(t)}\alpha_j^{(t)}$.

5.4 Inverse gamma distribution

Let $\{X_i\}_{i=1}^n \overset{\text{iid}}{\sim} \text{IGamma}(\alpha, \beta)$ with pdf $\beta^\alpha\Gamma^{-1}(\alpha)x^{-\alpha-1}\exp(-\beta/x)$ for $x > 0$, where $\alpha\,(> 0)$ and $\beta\,(> 0)$ are two parameters. The log-likelihood function is $\ell(\alpha, \beta) = -nG(\boldsymbol{x})(\alpha + 1) - \beta\sum_{i=1}^n x_i^{-1} + n\left[\alpha\log\beta - \log\Gamma(\alpha)\right]$, where $G(\boldsymbol{x}) \triangleq (1/n)\sum_{i=1}^n \log x_i$. Given α, we have $\beta = n\alpha/\sum_{i=1}^n x_i^{-1}$. Given β, we have

$$\ell(\alpha) \triangleq \ell(\alpha|\beta) = -nG(\boldsymbol{x})\alpha + n[\alpha\log\beta - \log\Gamma(\alpha)] + \text{constant},$$

and its first– and second–order derivatives are

$$\begin{aligned}
\ell'(\alpha) &= -nG(\boldsymbol{x}) + n\left[\log\beta - \psi(\alpha)\right] \quad \text{and} \\
\ell''(\alpha) &= -n\psi'(\alpha) \overset{(5.1)}{=} -n\sum_{m=0}^{\infty}\frac{1}{(m+\alpha)^2} = -n\sum_{m=0}^{m_0}\frac{1}{(m+\alpha)^2} - n\sum_{m=m_0+1}^{\infty}\frac{1}{(m+\alpha)^2} \\
&> -n\sum_{m=0}^{m_0}\frac{1}{(m+\alpha)^2} - \varepsilon \overset{(B.3)}{\geqslant} -n\sum_{m=0}^{m_0}\left[\frac{(1-r_m^{(t)})^3}{m^2} + \frac{r_m^{(t)3}}{\alpha^2}\right] - \varepsilon = -\frac{a_{23}^{(t)}}{\alpha^2} - a_{24}^{(t)} - \varepsilon,
\end{aligned}$$

where the equality holds iff $\theta = \theta^{(t)}$, and here $\varepsilon \triangleq n\sum_{m=m_0+1}^{\infty} m^{-2}$,

$$r_m^{(t)} \triangleq \frac{\alpha^{(t)}}{m + \alpha^{(t)}}, \quad a_{23}^{(t)} = n\sum_{m=0}^{m_0} r_m^{(t)3} \quad \text{and} \quad a_{24}^{(t)} = \sum_{m=0}^{m_0}\frac{n(1-r_m^{(t)})^3}{m^2}.$$

Thus, the sub-optimal SeLF iteration is

$$\alpha^{(t+1)} = \operatorname{sol}\left\{(a_{24}^{(t)} + \varepsilon)\alpha^2 + a_{25}^{(t)}\alpha - a_{23}^{(t)} = 0,\ \forall\,\alpha, \alpha^{(t)} > 0\right\},$$

where $a_{25}^{(t)} \triangleq a_{23}^{(t)}/\alpha^{(t)} - (a_{24}^{(t)} + \varepsilon)\alpha^{(t)} - \ell'(\alpha^{(t)})$.

6. Convergence properties

To analysis the convergence rate for the SeLF algorithm, we define $\theta^{(t+1)} \triangleq h(\theta^{(t)})$ as the $(t+1)$-th approximation of the MLE $\hat{\theta}$. Theorem 3 below states that the SeLF iteration (2.9) has a linear convergence rate. If $b(\theta|\theta^{(t)})$ is the optimal SeLF of $\ell(\theta)$ at $\theta = \theta^{(t)}$ and there exists a closed–form solution to the maximization/equation in (2.9), Theorem 4 states that the convergence rate of the optimal SeLF algorithm is quadratic.

Theorem 3 (Linear convergence rate for the SeLF iteration). Let the SeLF $b(\theta)$ be defined by (2.2), and the $Q(\theta|\theta^{(t)})$ function be given by (2.8). Then the SeLF iteration

$$\theta^{(t+1)} = \text{sol}\left\{\ell'(\theta^{(t)}) + \int_{\theta^{(t)}}^{\theta} b(z)\,\mathrm{d}z = 0,\ \forall\,\theta, \theta^{(t)} \in \Theta\right\} \triangleq h(\theta^{(t)}) \tag{6.1}$$

holds a linear convergence rate given by

$$r_1 = \lim_{t\to\infty} \frac{|\theta^{(t+1)} - \hat{\theta}|}{|\theta^{(t)} - \hat{\theta}|} = |h'(\hat{\theta})| = 1 - \frac{\ell''(\hat{\theta})}{b(\hat{\theta})} \in [0, 1). \qquad \|$$

Theorem 4 (Quadratic convergence rate for the optimal SeLF iteration). Let $b(\theta|\theta^{(t)})$ be an optimal SeLF of $\ell(\theta)$ at $\theta = \theta^{(t)}$ satisfying (4.1)–(4.2). Then the optimal SeLF iteration

$$\theta^{(t+1)} \;=\; \text{sol}\left\{\ell'(\theta^{(t)}) + \int_{\theta^{(t)}}^{\theta} b(z|\theta^{(t)})\,\mathrm{d}z = 0,\ \forall\,\theta^{(t)}, \theta \in \Theta\right\} \triangleq h(\theta^{(t)}) \tag{6.2}$$

has a quadratic convergence rate given by

$$r_2 = \lim_{t\to\infty} \frac{|\theta^{(t+1)} - \hat{\theta}|}{|\theta^{(t)} - \hat{\theta}|^2} = \frac{|h''(\hat{\theta})|}{2} = \left|\frac{b_1'(\hat{\theta}, \hat{\theta}) + 2b_2'(\hat{\theta}, \hat{\theta}) - \ell'''(\hat{\theta})}{2\ell''(\hat{\theta})}\right|,$$

where

$$B(\theta|\theta^{(t)}) \;\triangleq\; \int_{\theta^{(t)}}^{\theta} b(z, \theta^{(t)})\,\mathrm{d}z,$$

where $b(z, \theta^{(t)}) \triangleq b(z|\theta^{(t)})$. $\|$

7. Numerical experiments

7.1 Comparisons of SeLF algorithm and Newton's method

To compare the SeLF algorithm and the Newton's method, we consider to use the logarithmic series distribution and Weibull distribution as examples to illustrate the advantage of strongly stably convergence property of the SeLF algorithm. In both cases, we choose the sample sizes $n = 400$ and the other parameters are selected as follows:

(A1) Logarithmic series distribution. The true values of β are set to be 0.5, 1, 2, 5.

(B1) Weibull distribution. The true values of $\{\theta, \lambda\}$ are set to be $\{0.5, 0.5\}$, $\{1, 1\}$, $\{2, 0.5\}$, $\{3, 1\}$.

For a given $\{n, \beta\}$, we generate $\{y_i\}_{i=1}^n \overset{\text{iid}}{\sim} \text{Logseries}(\beta)$ by the inversion method (Seber & Salehi 2012) based on its pmf. Then, we use the SeLF algorithm (3.4) and Newton's method to calculate the MLE $\{\hat{\beta}\}$ with the generated $\{y_i\}_{i=1}^n$ and initial values randomly chosen from $U(1, 5)$. Finally, we independently repeat this process 10,000 times. The percentage of converged algorithms among 10,000 repetitions (denoted by Percentage), the average number of iterations for converged algorithms among 10,000 repetitions (denoted by N), and the total computation time in second of calculating $\{\hat{\beta}\}$ for 10,000 repetitions [denoted by Time(s)] are recorded and displayed in Table 1.

For a given $\{n, \theta, \lambda\}$, we generate $\{y_i\}_{i=1}^n \overset{\text{iid}}{\sim} \text{Weibull}(\theta, \lambda)$ by the inversion method (Seber & Salehi 2012) based on its pmf. Similarly, we can calculate the MLEs $\{\hat{\theta}, \hat{\lambda}\}$ through the SeLF algorithm (3.6) and Newton's method with the generated $\{y_i\}_{i=1}^n$ and initial values randomly chosen from $\theta^{(0)} \overset{\text{iid}}{\sim} U(1, 5)$ for 10,000 repetitions. The values of Percentage, N and Time(s) are reported in Table 2.

Table 1. Comparisons of two algorithms in terms of the percentage of converged algorithms, the average number of iterations, and the total computation time in second in (A1)

Parameters	$\beta = 0.5$			$\beta = 1$		
Algorithms	SeLF	F–SeLF	Newton	SeLF	F–SeLF	Newton
Percentage	100.00%	100.00%	**16.42%**	100.00%	100.00%	**32.21%**
N	41.35	19.75	5.96	30.83	13.85	6.08
Time(s)	150.84	143.29	251.43	158.65	153.49	234.48
Parameters	$\beta = 2$			$\beta = 5$		
Algorithms	SeLF	F–SeLF	Newton	SeLF	F–SeLF	Newton
Percentage	100.00%	100.00%	**61.49%**	100.00%	100.00%	**99.99%**
N	25.18	10.72	6.24	22.83	9.82	6.72
Time(s)	187.57	183.93	207.87	268.71	264.69	261.37

Note: SeLF denotes the SeLF iteration in (3.4); F–SeLF denotes the fast SeLF iteration; Percentage is the percentage of converged algorithms among 10,000 repetitions for different initial values randomly chosen from $U(0, 5)$; N is the average number of iterations for converged algorithms among 10,000 repetitions; Time(s) is the total computation time in second of calculating $\hat{\beta}$ for 10,000 repetitions.

Table 2. Comparisons of two algorithms in terms of the percentage of converged algorithms, the average number of iterations, and the total computation time in second in (B1)

Parameters	$\{\theta, \lambda\} = \{0.5, 0.5\}$			$\{\theta, \lambda\} = \{1, 1\}$		
Algorithms	SeLF	F–SeLF	Newton	SeLF	F–SeLF	Newton
Percentage	100.00%	100.00%	**93.70%**	100.00%	100.00%	**99.91%**
N	103.82	50.77	6.49	70.94	35.06	5.99
Time(s)	176.57	98.83	11.19	121.70	69.80	10.75
Parameters	$\{\theta, \lambda\} = \{2, 0.5\}$			$\{\theta, \lambda\} = \{3, 1\}$		
Algorithms	SeLF	F–SeLF	Newton	SeLF	F–SeLF	Newton
Percentage	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%
N	56.21	27.41	6.44	47.81	23.37	6.38
Time(s)	97.39	55.13	11.49	86.10	49.63	12.79

Note: SeLF denotes the SeLF iteration in (3.4); F–SeLF denotes the fast SeLF iteration; Percentage is the percentage of converged algorithms among 10,000 repetitions for different initial values randomly chosen from $U(0, 5)$; N is the average number of iterations for converged algorithms among 10,000 repetitions; Time(s) is the total computation time in second of calculating $\hat{\beta}$ for 10,000 repetitions.

7.2 Computation efficiency for the optimal and Second-stage optimal SeLF algorithms

(A2) YuleC-Simon distribution. The true values of β are set to be 0.5, 1, 5, 10.

(B2) Inverse gamma distribution. The true values of $\{\theta, \lambda\}$ are set to be $\{0.3, 0.3\}$, $\{0/5, 0.5\}$, $\{1, 1\}$, $\{2, 0.5\}$.

Table 3. Comparisons of five algorithms in terms of the percentage of converged algorithms, the average number of iterations, and the total computation time in second for Yule–Simon distribution

Parameters	$\theta = 0.5$			
Algorithms	F–US	Newton	F–fixed–point	O–SeLF
Percentage	100.00%	**71.68%**	100.00%	100.00%
N	5.219	10.584	6.253	**4.683**
Time(s)	**68.000**	4624.439	81.214	126.640
Parameters	$\theta = 1$			
Algorithms	F–US	Newton	F–fixed–point	O–SeLF
Percentage	100.00%	**69.44%**	100.00%	100.00%
N	5.716	6.970	6.462	**4.592**
Time(s)	**44.782**	1930.883	50.745	**46.899**
Parameters	$\theta = 5$			
Algorithms	F–US	Newton	F–fixed–point	O–SeLF
Percentage	100.00%	100.00%	100.00%	100.00%
N	5.467	6.746	38.864	**4.371**
Time(s)	34.323	41.407	238.848	**30.737**
Parameters	$\theta = 10$			
Algorithms	F–US	Newton	F–fixed–point	O–SeLF
Percentage	100.00%	100.00%	100.00%	100.00%
N	6.658	8.972	88.569	**4.877**
Time(s)	40.029	54.522	522.512	**34.368**

Note: F–US denotes the fast US algorithm; F–fixed–point denotes the fast fixed–point algorithm; O–SeLF denotes the optimal SeLF algorithm defined in (5.2). Percentage is the percentage of converged algorithms among 10,000 repetitions for different initial values randomly chosen from $U(1, 5)$; N is the average number of iterations for converged algorithms among 10,000 repetitions; Time(s) is the total computation time in second of calculating $\hat{\theta}$ for 10,000 repetitions.

Table 4. Comparisons of two algorithms in terms of the percentage of converged algorithms, the average number of iterations, and the total computation time in second in (B1)

Parameters	$\{\theta, \lambda\} = \{0.3, 0.3\}$			$\{\theta, \lambda\} = \{0.5, 0.5\}$		
Algorithms	S–SeLF	F–SeLF	Newton	S–SeLF	F–SeLF	Newton
Percentage	100.00%	100.00%	**19.51%**	100.00%	100.00%	**79.69%**
N	9.222	9.090	8.624	12.442	10.638	10.870
Time(s)	2.584	2.613	1.024	3.251	3.015	2.245
Parameters	$\{\theta, \lambda\} = \{1, 1\}$			$\{\theta, \lambda\} = \{2, 0.5\}$		
Algorithms	S–SeLF	F–SeLF	Newton	S–SeLF	F–SeLF	Newton
Percentage	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%
N	21.030	17.967	17.841	50.663	37.292	37.450
Time(s)	4.760	4.387	3.764	10.656	8.447	6.799

Note: S–SeLF denotes the sub-optimal SeLF iteration in subsection 4.2; F–SeLF denotes the fast SeLF algorithm; Percentage is the percentage of converged algorithms among 10,000 repetitions for different initial values randomly chosen from $U(0, 5)$; N is the average number of iterations for converged algorithms among 10,000 repetitions; Time(s) is the total computation time in second of calculating $\hat{\beta}$ for 10,000 repetitions.

8. Discussion

References

Balakrishnan N and Kundu D (2019). Birnbaum–Saunders distribution: A review of models, analysis, and applications. *Applied Stochastic Models in Business and Industry* **35**(1): 4–49.

Burr IW (1942). Cumulative frequency functions. *The Annals of Mathematical Statistics* **13**(2): 215–232.

Engelhardt M, Bain LJ and Wright FT (1981). Inferences on the parameters of the Birnbaum–Saunders fatigue life distribution based on maximum likelihood estimation. *Technometrics* **23**(3): 251–256.

Gorgi P (2020). BetaCnegative binomial auto-regressions for modelling integer-valued time series with extreme observations. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)* **82**(5): 1325–1347.

Jaakkola TS and Jordan MI. (2000). Bayesian parameter estimation via variational methods. *Statistics and Computing* **10**(1): 25–37.

Leiva V (2015). *The Birnbaum–Saunders Distribution.* London: Academic Press.

Patil GP (1962). Some methods of estimation for the logarithmic series distribution. *Biometrics* **18**(1): 68–75.

Seber GAF and Salehi MM (2012). *Adaptive Sampling Designs: Inference for Sparse and Clustered Populations*, Chapter 5: Inverse sampling methods. New York: Springer (Springer Briefs in Statistics Book Series).

Wang Z (2011). One mixed negative binomial distribution with application. *Journal of Statistical Planning and Inference* **141**(3): 1153–1160.

Wingo DR (1983). Maximum likelihood methods for fitting the Burr type XII distribution to life test data. *Biometrical Journal* **25**(1): 77–84.

Wingo DR (1993). Maximum likelihood methods for fitting the Burr type XII distribution to multiply (progressively) censored life test data. *Metrika* **40**(1): 203–210.

Appendix A: Proofs of Theorem 1–3

Proof of Theorem 1. (i) We first prove that the SeLF algorithm strongly stably converges to the maximizer $\hat{\theta}$. Let $\theta^{(t)}$ be the t-th iteration of $\hat{\theta}$, then we only need to prove that

$$\theta^{(t)} < \theta^{(t+1)} \leqslant \hat{\theta}, \quad \text{if} \quad \theta^{(t)} < \hat{\theta}, \tag{A.1}$$

$$\hat{\theta} \leqslant \theta^{(t+1)} < \theta^{(t)}, \quad \text{if} \quad \theta^{(t)} > \hat{\theta}. \tag{A.2}$$

For the case of $\theta^{(t)} < \hat{\theta}$, note that

$$\ell'(\theta^{(t)}) + \int_{\theta^{(t)}}^{\theta^{(t+1)}} \ell''(z)\,\mathrm{d}z \overset{(2.2)}{\geqslant} \ell'(\theta^{(t)}) + \int_{\theta^{(t)}}^{\theta^{(t+1)}} b(z)\,\mathrm{d}z \;=\; 0,$$

we have

$$\int_{\theta^{(t)}}^{\theta^{(t+1)}} [\ell''(z) - b(z)]\,\mathrm{d}z \;\geqslant\; 0,$$

implying $\theta^{(t)} < \theta^{(t+1)}$ because $\ell''(\theta) - b(\theta) > 0$ for all $\theta \in \mathbf{\Theta}$. From

$$\begin{aligned} 0 \;&=\; \ell'(\theta^{(t)}) + \int_{\theta^{(t)}}^{\hat{\theta}} \ell''(z)\,\mathrm{d}z \;=\; \ell'(\theta^{(t)}) + \int_{\theta^{(t)}}^{\theta^{(t+1)}} \ell''(z)\,\mathrm{d}z + \int_{\theta^{(t+1)}}^{\hat{\theta}} \ell''(z)\,\mathrm{d}z \\ &\overset{(2.2)}{\geqslant} \ell'(\theta^{(t)}) + \int_{\theta^{(t)}}^{\theta^{(t+1)}} \ell''(z)\,\mathrm{d}z + \int_{\theta^{(t+1)}}^{\hat{\theta}} b(z)\,\mathrm{d}z, \end{aligned}$$

we have

$$\int_{\theta^{(t+1)}}^{\hat{\theta}} \ell''(z)\,\mathrm{d}z - \int_{\theta^{(t+1)}}^{\hat{\theta}} b(z)\,\mathrm{d}z \;=\; \int_{\theta^{(t+1)}}^{\hat{\theta}} [\ell''(z) - b(z)]\,\mathrm{d}z \;\geqslant\; 0,$$

implying $\theta^{(t+1)} < \hat{\theta}$ because $\ell''(\theta) - b(\theta) > 0$ for all $\theta \in \mathbf{\Theta}$. So (A.1) is true. For the case of $\theta^{(t)} > \hat{\theta}$, similarly, we can prove (A.2).

(ii) From (A.1) and (A.2), we can see that the SeLF algorithm does not depend on any initial values in Θ. □

Proof of Theorem 3. From (6.1), we have

$$\ell'(\theta^{(t)}) + \int_{\theta^{(t)}}^{h(\theta^{(t)})} b(z)\,\mathrm{d}z = 0. \tag{A.3}$$

By taking the first-derivative about $\theta^{(t)}$ on the both sides of (A.3), we obtain

$$\ell''(\theta^{(t)}) + b(h(\theta^{(t)}))h'(\theta^{(t)}) - b(\theta^{(t)}) = 0.$$

By letting $t \to \infty$, we have

$$h'(\hat{\theta}) = 1 - \frac{\ell''(\hat{\theta})}{b(\hat{\theta})} \in (0, 1),$$

because $\ell''(\hat{\theta}) < 0$ and $\ell''(\hat{\theta}) > b(\hat{\theta})$. □

Proof of Theorem 4. From (6.2), we have

$$\ell'(\theta^{(t)}) + \int_{\theta^{(t)}}^{h(\theta^{(t)})} b(z, \theta^{(t)})\, \mathrm{d}z = 0. \tag{A.4}$$

By taking the first-derivative about $\theta^{(t)}$ on the both sides of (A.4), we obtain

$$\ell''(\theta^{(t)}) + b(h(\theta^{(t)}), \theta^{(t)})h'(\theta^{(t)}) - b(\theta^{(t)}, \theta^{(t)}) + \int_{\theta^{(t)}}^{h(\theta^{(t)})} b_2'(z, \theta^{(t)})\, \mathrm{d}z = 0,$$

where $b_i'(x_1, x_2) \triangleq \partial b(x_1, x_2)/\partial x_i$ for $i = 1, 2$. By letting $t \to \infty$, we have

$$0 = \ell''(\hat{\theta}) + b(\hat{\theta}, \hat{\theta})h'(\hat{\theta}) - b(\hat{\theta}, \hat{\theta}) + \int_{\hat{\theta}}^{\hat{\theta}} b_2'(z, \hat{\theta})\, \mathrm{d}z = \ell''(\hat{\theta})h'(\hat{\theta}),$$

and then $h'(\hat{\theta}) = 0$. By taking the second-derivative about $\theta^{(t)}$ on the both sides of (A.4), we obtain

$$\begin{aligned} 0 \;=\; & \ell'''(\theta^{(t)}) + b_1'(h(\theta^{(t)}), \theta^{(t)})(h'(\theta^{(t)}))^2 + b_2'(h(\theta^{(t)}), \theta^{(t)})h'(\theta^{(t)}) + b(h(\theta^{(t)}), \theta^{(t)})h''(\theta^{(t)}) \\ & - b_1'(\theta^{(t)}, \theta^{(t)}) - b_2'(\theta^{(t)}, \theta^{(t)}) + b_2'(h(\theta^{(t)}), \theta^{(t)})h'(\theta^{(t)}) - b_2'(\theta^{(t)}, \theta^{(t)}) \\ & + \int_{\theta^{(t)}}^{h(\theta^{(t)})} b_{22}''(z, \theta^{(t)})\, \mathrm{d}z. \end{aligned}$$

where $b_{22}''(x_1, x_2) \triangleq \partial^2 b(x_1, x_2)/\partial x_2^2$. By letting $t \to \infty$, we have

$$\begin{aligned} 0 \;=\; & \ell'''(\hat{\theta}) + b_1'(\hat{\theta}, \hat{\theta})(h'(\hat{\theta}))^2 + b_2'(\hat{\theta}, \hat{\theta})h'(\hat{\theta}) + b(\hat{\theta}, \hat{\theta})h''(\hat{\theta}) - b_1'(\hat{\theta}, \hat{\theta}) \\ & - 2b_2'(\hat{\theta}, \hat{\theta}) + b_2'(\hat{\theta}, \hat{\theta})h'(\hat{\theta}) + \int_{\hat{\theta}}^{\hat{\theta}} b_{22}''(z, \hat{\theta})\, \mathrm{d}z. \end{aligned}$$

Thus, we have

$$\begin{aligned} 0 &= \ell'''(\hat{\theta}) - b_1'(\hat{\theta},\hat{\theta}) - 2b_2'(\hat{\theta},\hat{\theta}) + b(\hat{\theta},\hat{\theta})h''(\hat{\theta}) \\ &\overset{(4.2)}{=} \ell'''(\hat{\theta}) - b_1'(\hat{\theta},\hat{\theta}) - 2b_2'(\hat{\theta},\hat{\theta}) + \ell''(\hat{\theta})h''(\hat{\theta}) \end{aligned}$$

and

$$\frac{|\theta^{(t+1)} - \hat{\theta}|}{|\theta^{(t)} - \hat{\theta}|^2} = \frac{|h''(\hat{\theta})|}{2} = \left| \frac{b_1'(\hat{\theta},\hat{\theta}) + 2b_2'(\hat{\theta},\hat{\theta}) - \ell'''(\hat{\theta})}{2\ell''(\hat{\theta})} \right|.$$

indicating that the SeLF iteration with optimal SeLF has a quadratic convergence rate given by (6.2). □

Appendix B: Inequalities for constructing some SeLFs

B.1 Discrete version of Jensen's inequality

Let $f(z)$ is a *concave* function defined on a convex set $\mathbb{C}$, i.e., $f''(z) \leqslant 0$ for all $z \in \mathbb{C}$. The discrete version of Jensen's inequality is

$$f\left(\sum_{k=1}^{K} q_k z_k\right) \geqslant \sum_{k=1}^{K} q_k f(z_k), \tag{B.1}$$

which is true for any probability weights $\{q_k\}_{k=1}^K$ satisfying: $q_k > 0$ and $\sum_{k=1}^K q_k = 1$.

1) Let $u_1(\boldsymbol{\phi}) > 0$ and $u_2(\boldsymbol{\phi}) > 0$. In (B.1), set $K = 2$, $f(z) = \log(z)$ for $z > 0$, then we obtain

$$\begin{aligned} \log\left[u_1(\boldsymbol{\phi}) + u_2(\boldsymbol{\phi})\right] &\geqslant v(\boldsymbol{\phi}^{(t)}) \log\left[\frac{u_1(\boldsymbol{\phi})}{v(\boldsymbol{\phi}^{(t)})}\right] + [1 - v(\boldsymbol{\phi}^{(t)})] \log\left[\frac{u_2(\boldsymbol{\phi})}{1 - v(\boldsymbol{\phi}^{(t)})}\right] \\ &= v(\boldsymbol{\phi}^{(t)}) \log\left[u_1(\boldsymbol{\phi})\right] + [1 - v(\boldsymbol{\phi}^{(t)})] \log\left[u_2(\boldsymbol{\phi})\right] + c_0^{(t)}, \end{aligned} \tag{B.2}$$

where the equality holds iff $\boldsymbol{\phi} = \boldsymbol{\phi}^{(t)}$, and $c_0^{(t)}$ is a constant free from $\boldsymbol{\phi}$,

$$v(\boldsymbol{\phi}^{(t)}) \triangleq \frac{u_1(\boldsymbol{\phi}^{(t)})}{u_1(\boldsymbol{\phi}^{(t)}) + u_2(\boldsymbol{\phi}^{(t)})}.$$

2) Let $a \geqslant 0$ and $\theta > 0$. In (B.1), set $K = 2, f(z) = -z^{-2}$ for $z > 0$, then we obtain

$$-\frac{1}{(a+\theta)^2} = -\left[(1 - r^{(t)})\frac{a}{1 - r^{(t)}} + r^{(t)}\frac{\theta}{r^{(t)}}\right]^{-2} \overset{(B.1)}{\geqslant} -\left[\frac{(1 - r^{(t)})^3}{a^2} + \frac{r^{(t)3}}{\theta^2}\right], \tag{B.3}$$

where $r^{(t)} \triangleq \theta^{(t)}/(a+\theta^{(t)})$, $r^{(t)m} \triangleq (r^{(t)})^m$ for any positive integer m, and the equality holds iff $\theta = \theta^{(t)}$.

3) Let $a_i \geqslant 0$ for $i = 1, \ldots, n$ and $\theta > 0$. Note that $f(z) = z^{-2}$ is a convex function for $z > 0$, using (B.1) conversely, we have

$$\begin{aligned}\sum_{i=1}^{n}(a_i+\theta)^{-2} &= S^{(t)}\sum_{i=1}^{n}\frac{s_i^{(t)2}}{S^{(t)}}\left[s_i^{(t)}(a_i+\theta)\right]^{-2} \overset{\text{(B.1)}}{\geqslant} S^{(t)}\left[\sum_{i=1}^{n}\frac{s_i^{(t)3}}{S^{(t)}}(a_i+\theta)\right]^{-2} \\ &= S^{(t)3}\left[\sum_{i=1}^{n}a_i s_i^{(t)3}+\theta\sum_{i=1}^{n}s_i^{(t)3}\right]^{-2} = \frac{d_1}{(d_2+\theta)^2}, \end{aligned} \tag{B.4}$$

where the equality holds iff $\theta = \theta^{(t)}$, and

$$s_i^{(t)} \triangleq \frac{1}{a_i+\theta^{(t)}}, \quad S^{(t)} \triangleq \sum_{i=1}^{n}s_i^{(t)2}, \quad d_1 \triangleq \frac{S^{(t)3}}{(\sum_{i=1}^{n}s_i^{(t)3})^2}, \quad d_2 \triangleq \frac{\sum_{i=1}^{n}a_i s_i^{(t)3}}{\sum_{i=1}^{n}s_i^{(t)3}}.$$

B.2 Inequalities for exponential functions

4) Let $\theta \geqslant 0$, we have

$$-\mathrm{e}^{\theta} \geqslant -\frac{4\mathrm{e}^{2d_3}}{(2\theta^{(t)}-\theta)^2}, \quad \forall\, 0 \leqslant \theta < 2\theta^{(t)} \quad \text{and} \quad \theta^{(t)} > 0, \tag{B.5}$$

where $d_3 \triangleq \max(\theta^{(t)}-1, 0)$.

Proof of (B.5). The inequality in (B.5) is equivalent to the following inequality:

$$\mathrm{e}^{\theta/2}(2\theta^{(t)}-\theta) \leqslant 2\mathrm{e}^{d_3} = 2\mathrm{e}^{\theta^{(t)}-1}I(\theta^{(t)}>1)+2I(\theta^{(t)}\leqslant 1), \quad \forall\, 0 \leqslant \theta < 2\theta^{(t)}. \tag{B.6}$$

Define $g(\theta) \triangleq \mathrm{e}^{\theta/2}(2\theta^{(t)}-\theta)$. By setting $0 = g'(\theta) = 0.5\mathrm{e}^{\theta/2}(2\theta^{(t)}-\theta)-\mathrm{e}^{\theta/2}$, we obtain its root $\theta^* = 2(\theta^{(t)}-1)$. Thus $g(\theta) > 0$ if $\theta < \theta^*$ and $g(\theta) < 0$ if $\theta > \theta^*$. So, we have

$$\begin{aligned} g(\theta) &\leqslant g(\max(\theta^*, 0)) = \begin{cases} g(\theta^*) = 2\mathrm{e}^{\theta^{(t)}-1}, & \text{if } \theta^* > 0, \\ g(0) = 2\theta^{(t)}, & \text{if } \theta^* \leqslant 0 \end{cases} \\ &= \begin{cases} 2\mathrm{e}^{\theta^{(t)}-1}, & \text{if } \theta^{(t)} > 1, \\ 2\theta^{(t)}, & \text{if } \theta^{(t)} \leqslant 1 \end{cases} \\ &= 2\mathrm{e}^{\theta^{(t)}-1}I(\theta^{(t)}>1)+2\theta^{(t)}I(\theta^{(t)}\leqslant 1) \\ &\leqslant 2\mathrm{e}^{\theta^{(t)}-1}I(\theta^{(t)}>1)+2I(\theta^{(t)}\leqslant 1), \end{aligned}$$

which implies (B.6). $\square$

5) For any $\theta > 0$, we have

$$-\mathrm{e}^{-\theta} > -\frac{2}{3}\theta^{-2}. \tag{B.7}$$

Proof of (B.7). Define $g(\theta) \triangleq \mathrm{e}^{\theta} - 3\theta^2/2$ for $\theta \geqslant 0$. We only need to prove that $g(\theta) > 0$. Since $g'(\theta) = \mathrm{e}^{\theta} - 3\theta$, by letting $0 = g'(\theta)$, we have two positive roots, $\theta_1 = 0.6190613$ and $\theta_2 = 1.512135$. Thus, we know

$$g(\theta) \geqslant \min\{g(0),\ g(\theta_1),\ g(\theta_2)\} = \min(1,\ 1.282329,\ 1.106577) = 1 > 0. \qquad \square$$

B.3 A convex function

6) For any $tha > 0$,

$$h(\theta) = -\log\left[\exp(0.5\theta^{1/2}) + \exp(-0.5\theta^{1/2})\right] \tag{B.8}$$

is a convex function.

Proof of (B.8). The first– and second–order derivatives of $f(\theta)$ are given by

$$\begin{aligned} h'(\theta) &= -\frac{\exp(0.5\theta^{1/2}) - \exp(-0.5\theta^{1/2})}{4\theta^{1/2}\left[\exp(0.5\theta^{1/2}) + \exp(-0.5\theta^{1/2})\right]} \quad \text{and} \\ h''(\theta) &= \frac{1}{4}\frac{\mathrm{e}^{\theta^{\frac{1}{2}}}}{\theta^{3/2}\left(\mathrm{e}^{0.5\theta^{\frac{1}{2}}} + \mathrm{e}^{-0.5\theta^{\frac{1}{2}}}\right)^2} > 0. \end{aligned} \tag{B.9}$$

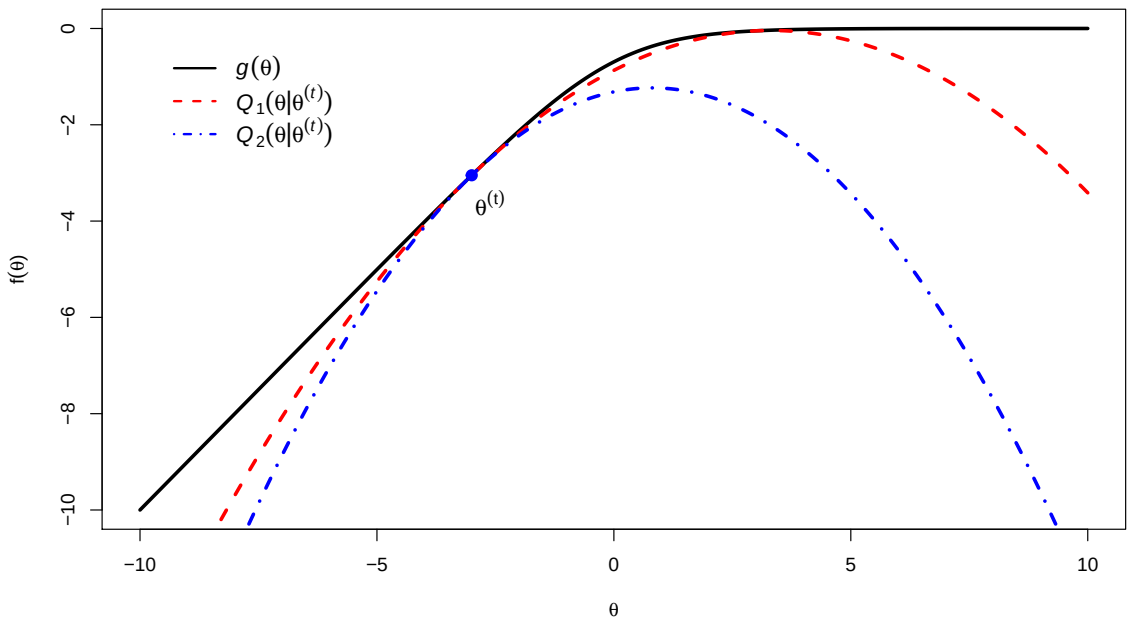
Since $f''(\theta) > 0$ for all $\theta \in \mathbb{R}^+$, we prove that $f(\theta)$ is a convex function. $\square$

To construct a Q function for $g(\theta) = -\log\left(\mathrm{e}^{\theta} + 1\right)$, Jaakkola & Jordan (2000) began by symmetrizing the log logistic function:

$$g(\theta) = \frac{\theta}{2} - \log\left(\mathrm{e}^{0.5\theta} + \mathrm{e}^{-0.5\theta}\right) = \frac{\theta}{2} - \log\left[\mathrm{e}^{0.5(\theta^2)^{\frac{1}{2}}} + \mathrm{e}^{-0.5(\theta^2)^{\frac{1}{2}}}\right],$$

and from (B.9), we know that $-\log\left(\mathrm{e}^{0.5\theta} + \mathrm{e}^{-0.5\theta}\right)$ is convex with respect to θ^2. By using the supporting hyperplane inequality, we have

$$g(\theta) \ \geqslant\ \frac{\theta}{2} + f(\theta^{2(t)}) + f'(\theta^{2(t)})(\theta^2 - \theta^{2(t)}) \ \triangleq\ Q_1(\theta|tht).$$



(a)

Figure 1. Plots of $g(\theta) = -\log\left(\mathrm{e}^{\theta/2} + \mathrm{e}^{-\theta/2}\right)$ and its two Q functions. The current iteration is $\theta^{(t)} = -3$ and the $Q_1(\theta|\theta^{(t)})$ and $Q_2(\theta|\theta^{(t)})$ are construct by using the inequality (3.5) and (B.9), respectively.

Figure 1 plots $g(\theta) = -\log\left(\mathrm{e}^{\theta/2} + \mathrm{e}^{-\theta/2}\right)$ and its two Q functions. The current iteration is $\theta^{(t)} = -3$ and the $Q_1(\theta|\theta^{(t)})$ and $Q_2(\theta|\theta^{(t)})$ are construct by using the inequality (3.5) and (B.9), respectively.